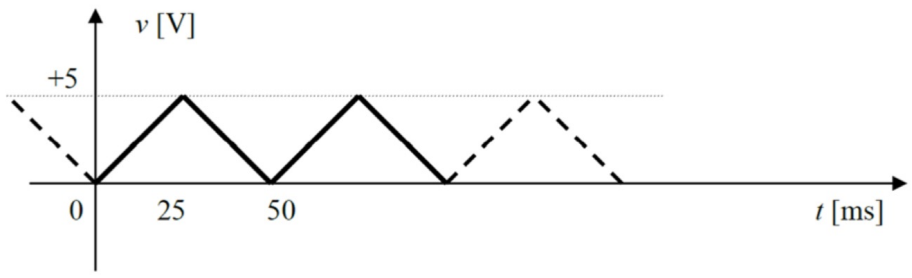
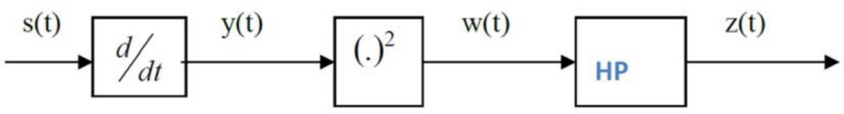
Exercise 2

Given the triangular periodic signal s(t) shown in the graph below, in which the times are expressed in milliseconds and the voltages in volts:



BETTER

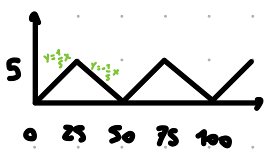
The signal s(t) enters a derivative block with respect to time, and the result y(t) enters a second block that performs the squared elevation, returning w(t).
Finally, the signal w(t) passes through an ideal high-pass (HP) filter, with cut-off frequency f_T=100 Hz, which returns z(t).



Write the expression of the coefficients cn of the Fourier series associated with the signals y(t) and w(t).
Plot the waveform output from the high-pass filter and calculate its power.

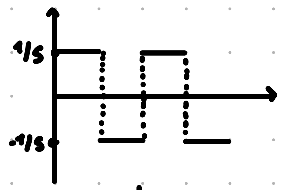
$T_0 : 50$ ms

s(t)



DERIVATIVE BLOCK

y(t)



FIND COEFFICIENTS

$$y(t)^{\uparrow} = \frac{1}{5} \text{RECT}\left(\frac{t - 25/2}{25}\right) \longrightarrow C_n^{y\uparrow} = \frac{1}{5} \cdot 25 \,\text{SINC}(f25)\, e^{-j2\pi f 25/2}$$

$$y(t)^{\downarrow} = -\frac{1}{5} \text{RECT}\left(\frac{t - 75/2}{25}\right) \longrightarrow C_n^{y\downarrow} = -\frac{1}{5}\, 25 \,\text{SINC}(f25)\, e^{-j2\pi f 75/2}$$

$$C_n^{y} = C_n^{y\uparrow} + C_n^{y\downarrow}$$

SQUARE BLOCK

w(t)

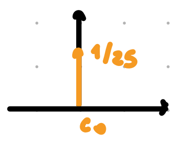
FIND COEFFICIENTS

$$w(t) = \frac{1}{25} \longrightarrow C_n = \frac{1}{25}$$

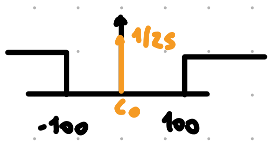
OR ALTERNATIVELY $C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t)\, e^{-\frac{j2\pi n t}{T_0}} dt = \dots = \frac{1}{25}$

FOURIER SERIES

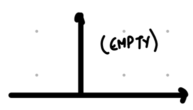
$C_n^{(w)}$



HIGH PASS FILTER



z(t)



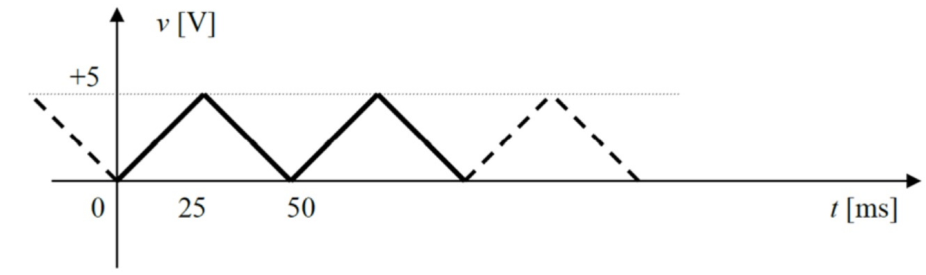
POWER

SIGNAL AREA IS 0

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} z(t)^2 dt = 0$$

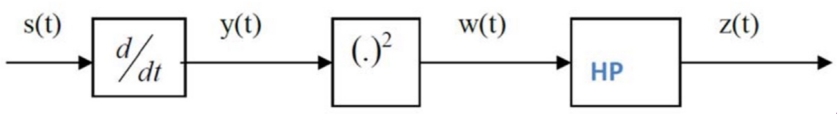
Exercise 2

Given the triangular periodic signal s(t) shown in the graph below, in which the times are expressed in milliseconds and the voltages in volts:



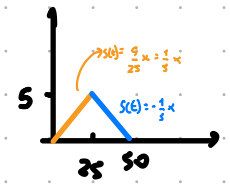
WORST

The signal s(t) enters a derivative block with respect to time, and the result y(t) enters a second block that performs the squared elevation, returning w(t).
Finally, the signal w(t) passes through an ideal high-pass (HP) filter, with cut-off frequency f_T=100 Hz, which returns z(t).



Write the expression of the coefficients cn of the Fourier series associated with the signals y(t) (1) and w(t) (2).
Plot the waveform (3) output from the high-pass filter and calculate its power (4).

(1)

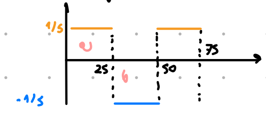


$$s(t) = \begin{cases} \frac{1}{5}x & \text{OTHERWISE} \\ -\frac{1}{5}x & x \in [25n \;\; 50n] \end{cases}$$

$$\boxed{T_0 = 50}$$

$$S(f) = A\left(\frac{BASE}{2}\right) sinc^2\left(f\, \frac{BASE}{2}\right) e^{-j2\pi f PEAK\_DELAY}$$

$$S(f) = 5 \cdot \frac{50}{2}\, sinc^2\left(f\, \frac{50}{2}\right) e^{-j2\pi f 25}$$



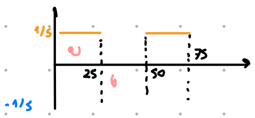
$$C_n = \frac{1}{T_0} S\left(\frac{n}{T_0}\right) \longrightarrow C_n = \frac{1}{T_0} A\tau\, sinc\left(\frac{n\tau}{T_0}\right) e^{-j2\pi f t}$$

$$C_n^a = \frac{1}{5} \cdot \frac{1}{50} \cdot 25\, sinc\left(f\, \frac{25}{50}\right) e^{-j2\pi \frac{25/2}{50}}$$

$$C_n^b = -\frac{1}{5} \cdot \frac{1}{50}\, 25\, sinc\left(f\, \frac{25}{50}\right) e^{-j2\pi \frac{75/2}{50}}$$

(2)

$$w(t) = y(t)^2$$



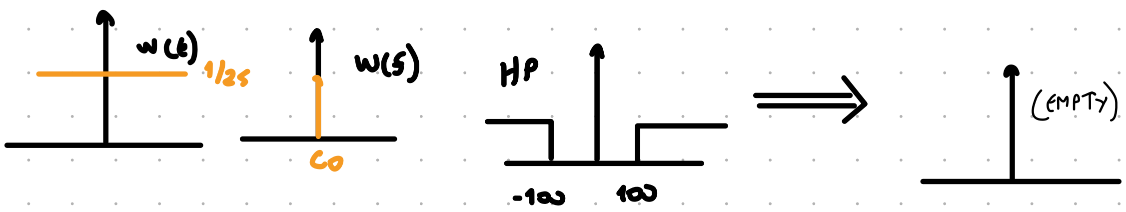
$\longrightarrow w(t) =$

NO CHANGE = NO FREQUENCY ⇒

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t)\, e^{-\frac{j2\pi n t}{T_0}}\, dt = \frac{1}{50} \int_{-25}^{25} \frac{1}{25} e^{-\frac{j2\pi n t}{T_0}}\, dt = \frac{1}{50} \cdot \frac{1}{25} \left[t\right]_{-25}^{25} = \frac{1}{50} \cdot \frac{1}{25} \cdot (25 - (-25)) = \frac{1}{25}$$

(3)

HIGH PASS FILTER WITH CUT-OFF FREQUENCY $f_T = 100Hz$



KEEP ONLY FREQUENCIES HIGHER THAN 100Hz, BUT WE ONLY HAVE 0Hz

(4)

AREA IS 0

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} z^2(t)\, dt = 0$$